COLOSSUS 3.0

Team Name: **VIPER**

Project / Idea Title: **PRAXIS - Multi-Agent Behavioral Assessment Protocol**

Track: **AI / Open Innovation / Future of Work**

Team Member 1(Lead Name): **Sai Vivek K N (Lead Architect) | Ph: 7892585383**

Team Member 2 Name: **Reehan Pasha (AI & Cloud Systems) | Ph: 8971183678**

Team Member 3 Name: **Kalpataru Institute of Technology (KIT), Tiptur**

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Team Member 4 Name: **VIPER Core Development Team (2-Person Sprint)**

The Problem - Why Does It Matter?

01 PROBLEM STATEMENT

02 WHO IS AFFECTED?

03 CURRENT PAIN POINTS

THE BREAKDOWN: GENERATIVE AI HAS DESTROYED STATIC INTERVIEWS

> 93% of LeetCode Medium/Hard questions & take-homes are solved instantly by Claude 3.5 & GPT-4o.
> Remote coding interviews suffer from widespread un-proctorable cheating (dual screens, background LLMs).
> Whiteboard trivia measures rote memorization and anxiety rather than real production engineering maturity.

WHO IS IMPACTED: ENGINEERING MANAGERS & CAPABLE BUILDERS

1. Tech Companies & Startups: Waste 42+ Senior Engineer hours per hire ($5,200 direct burn) screening false positives.
2. Non-Pedigree Candidates & Tier-2/3 Colleges: Brilliant self-taught builders are discarded by automated resume ATS filters purely due to lack of prestige branding.

CRITICAL PAIN POINTS & ECONOMIC METRICS

> $42,000 Average Cost of an engineering mis-hire in recruitment and onboarding loss.
> 46% of newly hired developers fail to navigate complex production codebases in their first 90 days.
> 60%+ surge in AI-hallucinated resumes and fraudulent remote candidate applications.
> The fundamental hiring signal is broken: The industry urgently requires Process Observation over rote trivia.

Idea / Solution

01 WHAT WE'RE BUILDING

02 HOW IT SOLVES IT

03 CORE VALUE PROPOSTION

WHAT WE'RE BUILDING: THE FLIGHT SIMULATOR FOR TALENT EVALUATION

Pilots don't take multiple-choice quizzes before flying; they fly high-fidelity simulators with engine failures. PRAXIS drops candidates into cloud sandboxes (Monaco IDE + E2B microVMs) to resolve authentic production incidents alongside autonomous AI co-workers.

HOW IT SOLVES IT: TRACKING 'PROCESS OVER ANSWERS' VIA REAL-TIME TELEMETRY

We evaluate HOW candidates think, not just what they write. The AI Observer captures keystroke flight-time variance, context switching, test-before-ship discipline, and ambiguity resolution under stakeholder pushback in real time.

CORE VALUE PROPOSITION: UNCHEATABLE EVALUATION & SOVEREIGN CREDENTIALS

> For Companies: 94% predictive hiring accuracy; 70% reduction in technical assessment time (sub-45 mins).
> For Candidates: Decentralized ZK-SNARK credentials proving skill across all employers without revealing proprietary code.
> Cheating Immunity: Behavioral micro-telemetry and procedural problem mutation make LLM copy-pasting mathematically ineffective.

How It Works

01 END-TO-END WORKFLOW

02 USER JOURNEY

03 TECHNOLOGY STACK

END-TO-END WORKFLOW: 6-STEP CLOSED-LOOP TALENT PIPELINE

1. Litmus Baseline Screen -> 2. Procedural ProblemBank Mutation -> 3. Interactive Sandbox Cockpit ->
4. AI Observer Telemetry Engine -> 5. Meta-Cognitive Evaluator (8 Dimensions) -> 6. SnarkJS ZK-SNARK Attestation.

DUAL USER JOURNEY: CANDIDATE & COMPANY PERSPECTIVES

> Candidate: Enters browser cockpit -> Diagnoses live rate-limiting race condition in Monaco IDE -> Interacts with AI PM (Alex) & Architect (Sam) -> Receives instant portable ZK competency passport.

> Company: Inspects candidate ranking by cognitive load score & approach quality -> Reviews telemetry playback -> Directly drafts proven top 1% talent via the Apprentice Market.

FULL-STACK PRODUCTION TECHNOLOGY STACK

> Cockpit Frontend: Next.js 16 (App Router), React 19, Monaco Editor, XTerm.js, Framer Motion.
> AI Agent Layer: Groq Llama-3.3-70B Versatile (<300ms ultra-low latency inference), Vercel AI SDK.
> Cloud Execution: E2B Cloud Sandboxes (Isolated Linux MicroVMs), Supabase PostgreSQL, Prisma ORM.
> Cryptographic Verification: SnarkJS (circom ZK-SNARKs), W3C Verifiable Credentials.

What Makes Us Different?

01 CORE FEATURES

02 EXISTING APPROACH

03 OUR INNOVATION / USP

01 CORE FEATURES: 4 PILLARS OF PRAXIS PLATFORM

> Procedural Mutation Engine: Mutates bug injection, edge cases, and parameters so no two tests leak.
> Low-Latency Multi-Agent Personas: Sub-300ms Groq agents roleplay PMs, Architects, and Clients with hidden constraints.
> Real-time Behavioral Telemetry: Millisecond tracking of flight-time variance, pause counts, and cognitive load score.
> ZK-SNARK Talent Credential: Cryptographically proven competency scores portable across employers.

02 EXISTING APPROACH: BROKEN BY DESIGN IN 2026

> HackerRank / LeetCode: Obsolete algorithm trivia; completely bypassed by LLMs; 0% real-world team correlation.
> Take-Home Projects: 8-12 hour candidate burn; rampant ChatGPT delegation and unobserved ghostwriting.
> One-Way Video Bots: Creepy facial recognition with zero technical evaluation depth and severe bias risks.

03 OUR INNOVATION & USP: THE 'PROCESS OVER ANSWERS' MOAT

> Live Cognitive Trajectory: We analyze HOW a candidate scopes, tests, and refactors under pressure.
> AI Co-Pilot Evaluation: Instead of banning AI, we evaluate how well the candidate guides and audits code.
> Cross-Domain Versatility: Single engine supports Engineering, Product Management, Enterprise Sales, and Data Science.



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Impact & Execution

01 EXPECTED IMPACT

02 BENEFICIARIES

03 SCALABILITY

04 IMPLEMENTATION ROADM

01 EXPECTED IMPACT & 02 BENEFICIARIES

> 70% Interview Time Savings: Compresses 3-4 screening rounds into a single sub-45-minute simulation.
> $3,200+ Saved Per Candidate Loop in Senior Engineering salary burn.
> Meritocratic Access: Empowers tier-2/3 college and self-taught developers with undeniable proof of execution ability.

03 SCALABILITY & BUSINESS VIABILITY

> Enterprise B2B SaaS: $499/month per engineering team for automated simulation pipelines.
> Apprentice Talent Market: 10% placement take-rate on direct verified talent drafts.
> Cloud MicroVM Infrastructure: Ephemeral Linux containers spin up on demand in <2 seconds with near-zero idle compute.

04 IMPLEMENTATION ROADMAP

> Phase 1 (Current / Hackathon MVP): Operational Next.js monorepo, Groq Llama-3.3-70B simulation engine, Monaco IDE, live telemetry observer, and company dashboard.
> Phase 2 (Months 1-3): Sub-200ms real-time voice agents, automated GitHub PR incident ingestion, and ATS integrations (Greenhouse, Lever, Ashby).
> Phase 3 (Months 4-6): Multi-agent collaborative team assessments & on-chain ZK talent registry.

Thank You

PRAXIS - PROVE YOUR PROCESS. BECOME VERIFIED.

TEAM VIPER | Kalpataru Institute of Technology (KIT), Tiptur

Sai Vivek K N (Lead Architect & Full-Stack) - Ph: 7892585383 | saivivek0707@gmail.com

Reehan Pasha (AI Systems & Infrastructure) - Ph: 8971183678 | reehanpasha1607@gmail.com